ZHANGYUE LEI

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EDUCATION

University of Science and Technology of China (USTC) *September 2025 to present*
B.Eng. in Computer Science, School of the Gifted Young College *Hefei, Anhui, China*
Huaxia Talented Program in Computer Science (founded and led by Prof. Hong An; 60 students selected from 2116 students of the whole grade)

Overall GPA: 3.74/4.30, 88.04/100; Major Ranking: 27/147
Core Courses: Fundamentals of Systems Programming (95, Top 10), Linear Algebra (*B1*) (90, Top 15)

National University of Singapore (NUS), School of Computing *May 2026 to July 2026*
Summer Workshop in Structure and Interpretation of Computer Programs *Singapore*

Completed the workshop covering programming paradigms, computational thinking, functional programming, recursive problem-solving, and the mathematical foundations of computer science through the SICP curriculum.

RESEARCH EXPERIENCE

1. Topic: ViT Compression for Medical Image Classification *April 2026*
RetinaMNIST Classification via Distillation-Guided Quantization

- Built a ViT-Small classifier for diabetic retinopathy grading on RetinaMNIST using knowledge distillation and GPTQ/AWQ low-bit quantization.
- Distilled a full-precision teacher into a quantized student, recovering accuracy under 4-bit compression with F1 above the baseline.
- Ranked among the leading teams in the 2nd China Algorithm Capability Competition National Final AI Track Model Lightweight Challenge.

2. Topic: NexOS Agent-OS *March 2026 to August 2026*
Agent-Native Kernel Extensions for a RISC-V Teaching OS

- Designed kernel Agents with capability identity, per-process Context, Tool Calls, and an event Loop for heartbeat, IPC, file events, and priority + aging.
- Won Second Prize in the National College Computer System Development Capability Competition.

3. Topic: CS229 Deep Learning Assignments *March 2026 to July 2026*
Dual-Framework Implementation in PyTorch (Python) and Burn (Rust)

- Implemented deep learning models in PyTorch (Python) and Burn (Rust), developing practical proficiency in both languages and frameworks.
- gained hands-on experience and explore the underlying principles of neural networks.

AWARDS AND HONORS

- Second Prize, National College Computer System Development Capability Competition (Operating System Design, Functional Challenge Track), National Level, China, Summer 2026
- Second Prize, China Algorithm Capability Competition (Final Contest), 2nd Session, First-Year Undergraduate Division, National Level, China, Spring 2026
- Second Prize, China Algorithm Capability Competition (Regional Contest), 2nd Session, First-Year Undergraduate Division, National Level, China, Fall 2025
- First Prize, 41st Chinese Physics Olympiad, Provincial Level, China, Senior 2, 2024
- Third Prize, Chinese Mathematical Olympiad (Preliminary Round), China, Senior 2, 2024

TECHNICAL STRENGTHS

English Fluency: TOEFL iBT 97 (Reading 25, Listening 29, Speaking 23, Writing 20)
Other Languages: Mandarin (native), Cantonese (fluent)
Computer Skills: C/C++, Python, Rust, TypeScript, LaTeX
Courses: CS229, Deep Learning; CS231n, Convolutional Neural Networks for Visual Recognition